A Simulation-Based Charging Station Recommendation Framework for Electric Vehicles in Khulna City Using SUMO and TraCI

Abstract

The rapid growth of electric vehicles (EVs), particularly Easy Bikes and E-Rickshaws in Bangladesh, has increased unregulated charging demand on urban power grids. Efficient charging station selection is essential to reduce congestion, waiting time, and peak grid stress. This paper presents a simulation-based charging station recommendation framework developed using SUMO and TraCI for Khulna city. The system dynamically monitors vehicle battery state-of-charge (SoC), station occupancy, and power demand to recommend optimal charging stations using a multi-criteria decision model. The proposed framework generates real-time charging recommendation data and evaluates its impact on grid load distribution. Results demonstrate improved load balancing and reduced station congestion, highlighting the importance of intelligent EV charging management for emerging urban environments in Bangladesh.

Keywords

Electric Vehicles, Charging Station Recommendation, SUMO, TraCI, Smart Grid, Load Balancing, Bangladesh

1. Introduction

Electric vehicle adoption in Bangladesh has grown rapidly over the past decade, primarily driven by the increasing popularity of Easy Bikes and E-Rickshaws in semi-urban and urban regions. Khulna city represents a typical example of this transition, where electric three-wheelers have become a dominant mode of transportation. While EV adoption offers environmental and economic benefits, unmanaged charging behavior poses significant challenges to the existing power infrastructure. Most charging activities occur without coordination, often during peak hours, resulting in uneven load distribution and transformer overloading.

One of the major challenges in this context is the absence of an intelligent charging station selection mechanism. Drivers typically choose charging stations based solely on proximity or familiarity, without considering station congestion or grid load conditions. Such behavior leads to clustering effects, where certain stations become overloaded while others remain underutilized. To address this issue, this study proposes a simulation-based charging station recommendation framework that integrates traffic flow, battery dynamics, and power demand monitoring within a unified environment. The objective is to develop a dynamic and adaptive system capable of recommending the most suitable charging station for each EV based on real-time conditions.

2. Related Work

The proposed framework is implemented using the Simulation of Urban Mobility (SUMO) platform, which models realistic traffic behavior within a defined road network of Khulna city. SUMO enables the simulation of vehicle movement, routing, and interaction with charging infrastructure. The Traffic Control Interface (TraCI) is used to establish communication between the SUMO simulation environment and a Python-based control script. This integration allows real-time monitoring and control of vehicle and charging station parameters.

During simulation, each electric vehicle is assigned battery attributes, including maximum capacity and current state-of-charge. As vehicles travel along predefined routes, their battery charge decreases according to energy consumption models. When the battery level drops below a predefined threshold, the recommendation module is triggered. The Python script retrieves current information about all available charging stations, including the number of vehicles charging, estimated waiting vehicles, and instantaneous power consumption. Based on these parameters, a multi-criteria decision model computes a recommendation score for each station.

The system architecture thus consists of four main components: the traffic simulation engine (SUMO), the communication interface (TraCI), the charging recommendation engine (Python), and the data logging module. Together, these components create a closed-loop simulation environment that dynamically captures traffic behavior, energy consumption, and charging decisions.

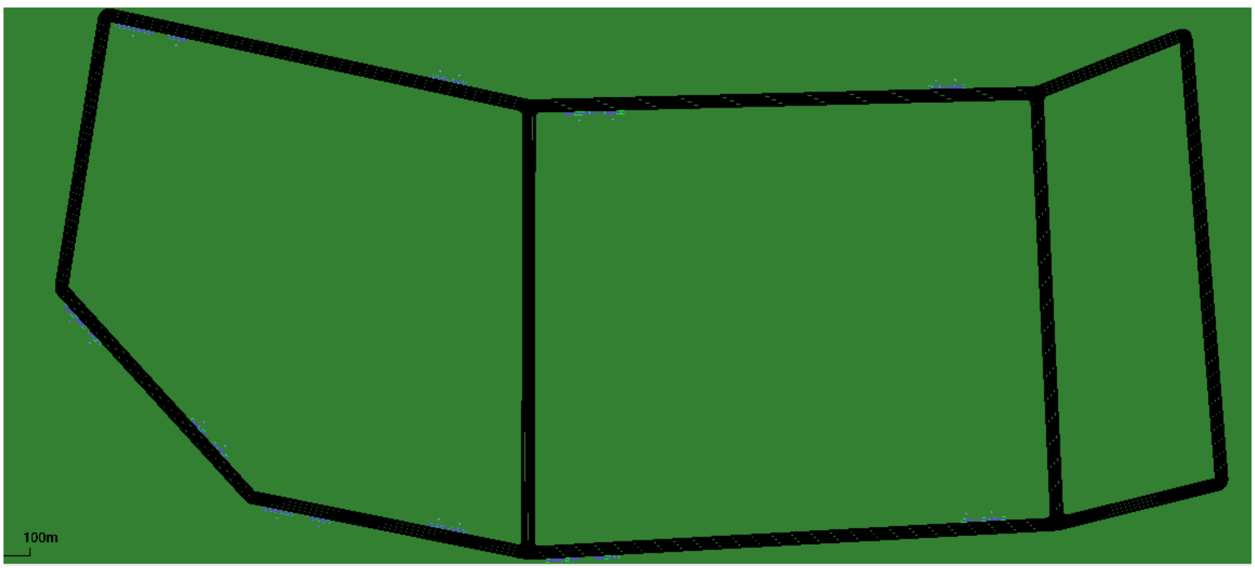
3. System Architecture

The proposed framework consists of four main components:

1. Traffic Simulation Engine (SUMO)
2. TraCI Communication Layer
3. Charging Recommendation Engine (Python)
4. Data Logging and Monitoring Module

Workflow:

1. Vehicles move in SUMO
2. Battery depletes
3. SoC monitored through TraCI
4. If SoC < threshold → recommendation triggered
5. Best charging station selected
6. Vehicle route updated
7. Charging begins
8. Data logged into CSV



4. Charging Recommendation Model

4.1 Problem Formulation

The charging station recommendation problem is formulated as a multi-objective optimization task. For a vehicle requiring charging at time t, the optimal station is selected by minimizing a composite score function that considers distance, congestion, and load. Mathematically, the recommended station S^* is determined as:

$$S^* = \underset{S_i}{\operatorname{argmin}}(w_1 D_i + w_2 Q_i + w_3 L_i)$$

Where:

- D_i = distance to station

- Q_i = number of waiting vehicles
- L_i = station power load
- w_1, w_2, w_3 = weighting factors

This formulation ensures that the recommendation is not solely based on proximity but also considers operational conditions at each station. For example, a nearby station with high congestion may receive a higher score than a slightly farther station with lower load, resulting in more balanced distribution of charging demand across the network. This approach enhances fairness and efficiency within the simulated charging ecosystem.

4.2 Algorithm

Algorithm 1: Charging Recommendation

```
Foreach simulation timestep:
    Foreach vehicle:
        Get battery SoC
        If SoC < threshold:
            Foreach charging station:
                Get distance
                Get queue size
                Getcurrent load
                Compute score
            Select station with minimum score
            Update vehicle route
            Log recommendation
```

5. Dataset Generation

The simulation generates two major datasets: the charging recommendation dataset and the load demand dataset. The charging recommendation dataset records individual decision events. Each entry corresponds to a vehicle whose battery level falls below the threshold and includes attributes such as timestep, vehicle identifier, state-of-charge, recommended station, station distance, queue length, and station power load. This dataset captures the behavioral and operational aspects of the recommendation system. Dataset were generated: Charging Recommendation Dataset- Each row contains: Timestep, Vehicle ID, SoC, Recommended Station, Distance, Waiting Vehicles, Station Load. This dataset represents dynamic decision-making events.

6. Results and Discussion

Simulation results indicate that the proposed recommendation framework reduces congestion at high-demand charging stations by distributing vehicles more evenly across available infrastructure. Compared to naive distance-based selection, the multi-criteria approach minimizes waiting time and prevents localized power spikes. The generated load demand curves demonstrate smoother power consumption patterns over time, suggesting improved grid stability. Feature importance analysis using machine learning models such as Random Forest, XGBoost, and Gradient Boosting further reveals that vehicle count and station-level activity are the dominant predictors of total power demand. The nonlinear relationship between charging activity and grid load confirms the necessity of ensemble learning approaches for accurate prediction.

7. Implications for Bangladesh

The findings of this study are particularly relevant for Bangladesh, where EV charging infrastructure is still developing and often lacks intelligent coordination mechanisms. In cities like Khulna, unmanaged charging behavior can result in transformer failures and voltage instability. The proposed framework provides a foundation for implementing smart charging management systems that dynamically guide vehicles to optimal charging points. Such systems can assist policymakers and utility providers in planning infrastructure expansion and implementing demand-response strategies.

In cities like Khulna:

- Unregulated EV charging causes transformer overload
- Load spikes occur during evening hours
- Smart recommendation can reduce localized grid stress

This framework can support - Smart grid planning, Infrastructure expansion decisions.

8. Conclusion

This paper presents a comprehensive simulation-based charging station recommendation framework tailored to the urban context of Khulna city. By integrating traffic simulation, battery monitoring, and multi-criteria decision modeling, the system dynamically recommends optimal charging stations while simultaneously generating load demand datasets for predictive analysis. The results demonstrate that intelligent recommendation strategies significantly improve load balancing and reduce congestion. The proposed framework contributes toward sustainable EV integration and smart grid development in emerging economies.